

Face Counter Using MATLAB

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This program helps count the number of persons present in a meeting hall or classroom at a time. The program logic can also be used in other applications, such as automatic temperature control which depends on the number of persons in a room. Video camera plays a very important role in this project.

If you try to count a large number of people in a hall, you may take time to count them and probably also make a mistake. With the help of this program, you can get the number of people instantly and more accurately using a camera and image processing.

Software program

A graphic user interface (GUI) allows users to perform tasks interactively through such controls as switches and sliders. You can create GUI and run it in MATLAB or as standalone application. The initial program output of this project is shown in Fig. 1 and its output after face detection is shown in Fig. 2.

Face detection algorithm. There are different types of algorithms used in face detection. Here, we used Viola-Jones algorithm for face detection with MATLAB program.

The algorithm actually works in following steps:

1. Creates a detector object using Viola-Jones algorithm
2. Takes the image from the video
3. Detects the features
4. Annotates the detected features

MATLAB functions. The program (testing.m) has many functions. Do not edit the functions as these are linkers

and non-executable codes. First of all, you have to find the format supported by the camera and its device ID using the command given below (also shown in Fig. 3):

```
info = imagehwinfo('winvideo')
640x480';
```

After knowing the device ID, you can change the device ID number in your source code.

We are having device ID as {1}, so have written '1' in the code, as mentioned below:

```
vid = videoinput('winvideo',1,'YUY2_640x480');
```

We also have other formats in MATLAB. You can check which format your camera supports by using commands below (also shown in Fig. 4):

```
info.DeviceInfo(1)
info.DeviceInfo.SupportedFormats
```

In Fig. 4, you can see that format 'YUY2_160x120' is the one which is

supported by the camera by default. But there are also other formats (resolutions) which your camera can support, as shown in last line in this screenshot. If you select another format and device number, you should make changes in the source code accordingly.

To detect a face or a particular feature on the faces of people, following steps are used in MATLAB program (testing.m):

1. Define and set up your cascade object detector using the constructor.

`detector=vision.CascadeObjectDetector`
It creates a system object detector that detects objects using the Viola-Jones algorithm. Its Classification Model property controls the type of object to detect. By default, the detector is configured to detect faces.

2. Call the step method with the input image I, the cascade object detector object, detector, points PTS and any optional properties. See the syntax below for using the step method. Use the step syntax with input image I, the selected cascade object detector object, and any optional properties to perform detection.

```
BBOX = step (detector, I)
```

It returns BBOX, an M-by-4 matrix defining M bounding boxes containing the detected objects. This method performs multi-scale object detection on the input image I. Each row of the output matrix BBOX contains a four-element vector (x, y, width and height) that specifies in pixels, the upper-left corner and size of a bounding box. Input image I must be a grayscale or true colour (RGB) image.

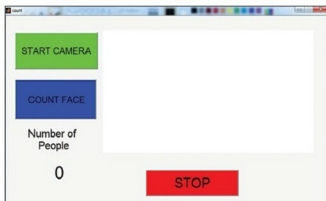


Fig. 1: Face-counter program output



Fig. 2: Face counter using MATLAB